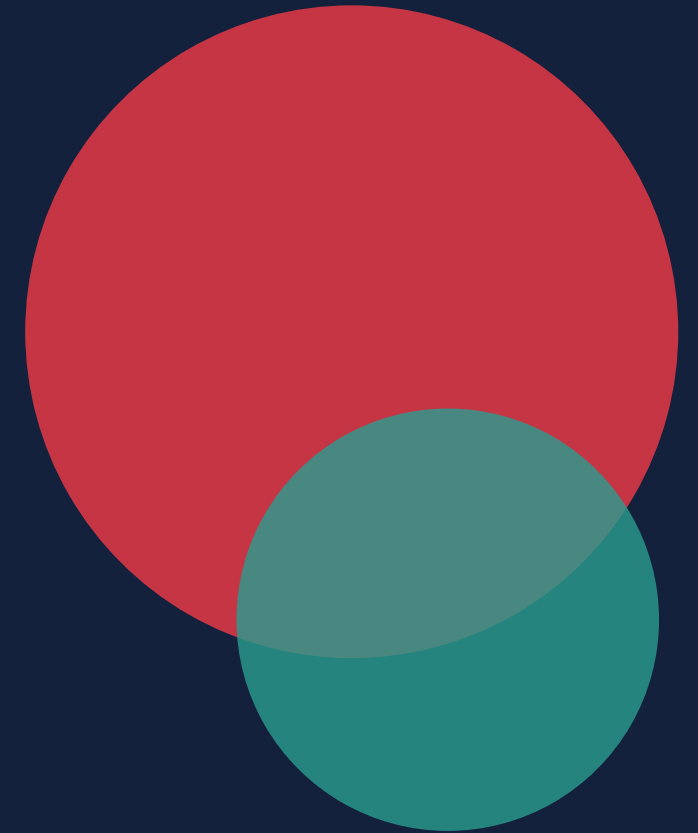


LectureLens

An offline, on-device AI classroom companion for Snapdragon-powered HP PCs

Listen • See • Understand • Translate • Ask — 100% on the NPU, no internet



The problem

Indian classrooms move fast, in one language, for students who think in many

1

Language gap

Many first-generation and regional-medium students struggle to follow fast English-medium lectures.

2

Notes vs. attention

Students must choose between writing notes and actually understanding the teacher.

3

Unreliable internet

Campus Wi-Fi and rural connectivity make cloud AI tools slow or unusable.

4

Privacy & cost

Recording classes to cloud services raises privacy concerns and needs paid subscriptions.

The solution: LectureLens

Turns every lecture into bilingual notes, flashcards and a quiz — privately, on the laptop

During class

- Live transcript of the teacher
- Board and slide text captured by webcam
- Key terms highlighted in real time

After class

- Structured notes + summary
- Tamil / Hindi translation in one click
- Flashcards and a 5-question quiz
- "Ask your lecture" Q&A

0

bytes of lecture data sent to the cloud

100%

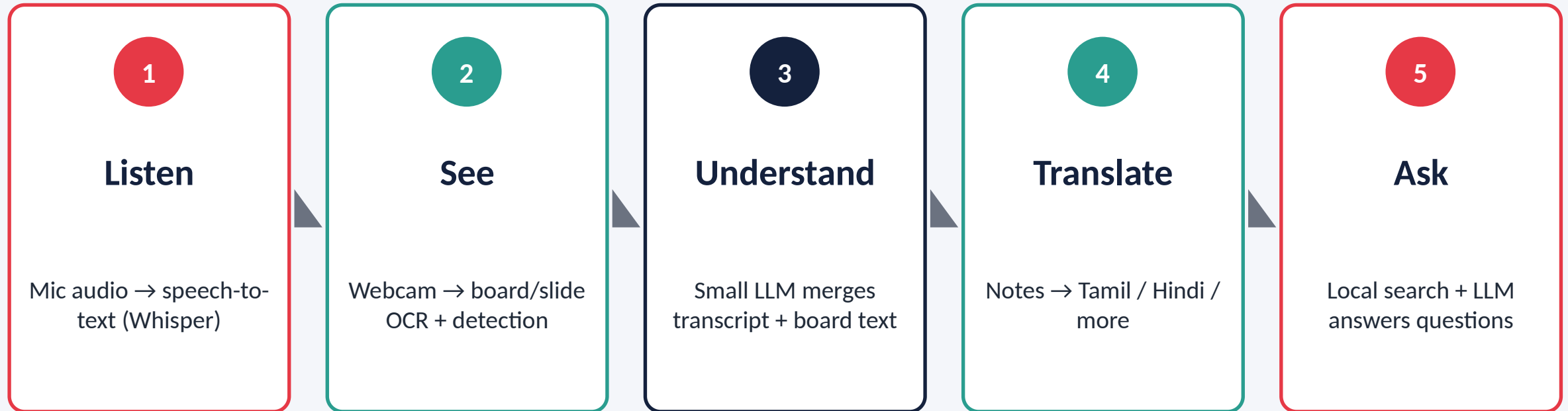
of AI inference on-device (NPU-first)

1-click

install — no account, no internet

How it works

Five on-device stages, orchestrated by one lightweight desktop app



Output: a searchable "lecture pack" per class — transcript, notes, translation, flashcards, quiz — saved locally.

AI models (Qualcomm AI Hub + open source)

Pre-optimized, quantized models compiled for the Snapdragon X-series Hexagon NPU

Task	Model	Runs on
Speech-to-text	Whisper (small / base) — Qualcomm AI Hub	NPU
Board / slide text	TrOCR or EasyOCR — Qualcomm AI Hub / open source	NPU
Board region detection	YOLO object detector — Qualcomm AI Hub	NPU
Notes, quiz, Q&A	Llama 3.2 3B / Phi-3.5-mini (INT4) — Qualcomm AI Hub	NPU
Translation	IndicTrans2 (AI4Bharat, open source)	NPU / CPU
Semantic search	MiniLM sentence embeddings (open source)	NPU / CPU

Runtime: ONNX Runtime with the QNN Execution Provider; models profiled and compiled for target devices through Qualcomm AI Hub.

Why Snapdragon-powered HP PCs

Dedicated NPU

The Hexagon NPU runs speech, vision and language models continuously while the CPU and GPU stay free for the student.

All-day battery

NPU inference is far more power-efficient than CPU/GPU, so LectureLens can run through a full day of classes.

Private by design

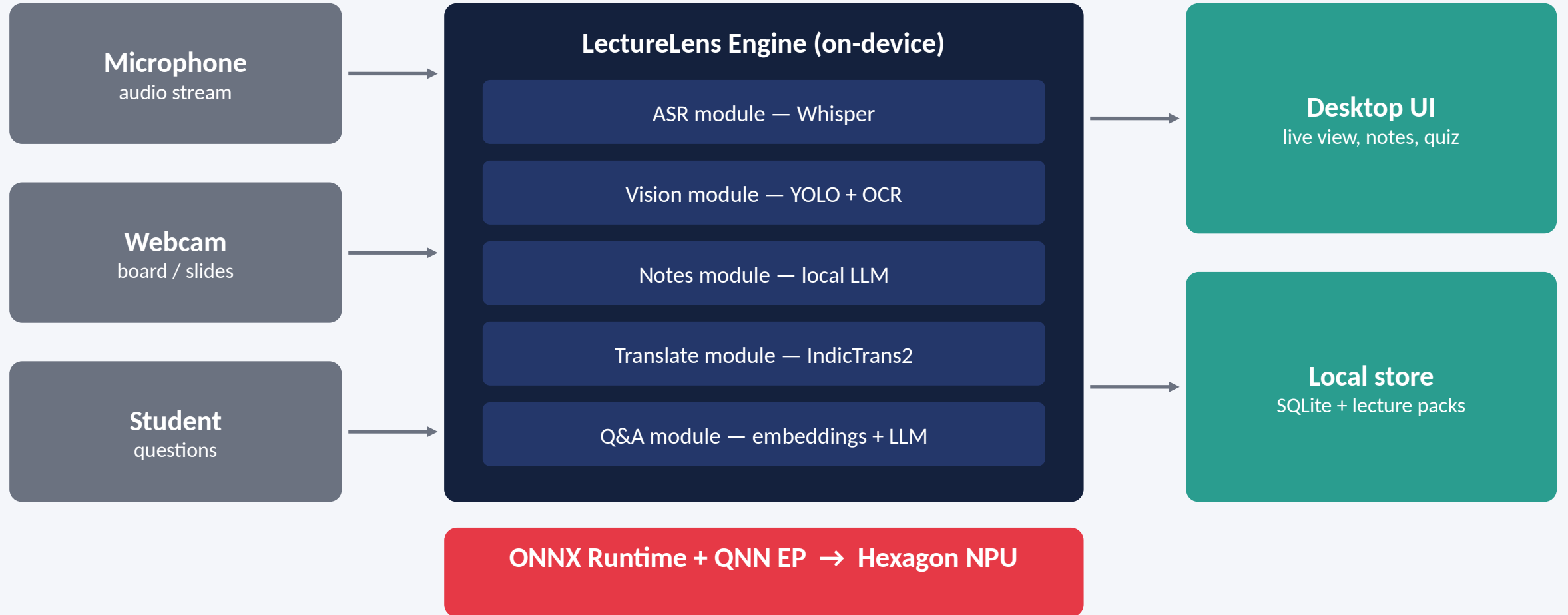
Everything is processed locally — lecture audio and notes never leave the laptop.

Works anywhere

Fully offline: campus, hostel, train or rural college with no connectivity.

Architecture

Modular Python engine with a desktop UI, packaged as a Windows on ARM (ARM64) app



Deployment & accessibility

Built so any student can install it in a minute and use it without help

One-click installer

Signed Windows ARM64 installer with models bundled — no Python, account or internet setup.

Offline-first

All features work with Wi-Fi off; optional sync only if the student chooses.

Multilingual UI

Interface and notes in English, Tamil and Hindi, extendable to more Indian languages.

Inclusive design

Large-text mode, high contrast, keyboard navigation and text-to-speech of notes for visually impaired students.

Hearing support

Live captions on screen help students with hearing difficulties follow the lecture.

Open source

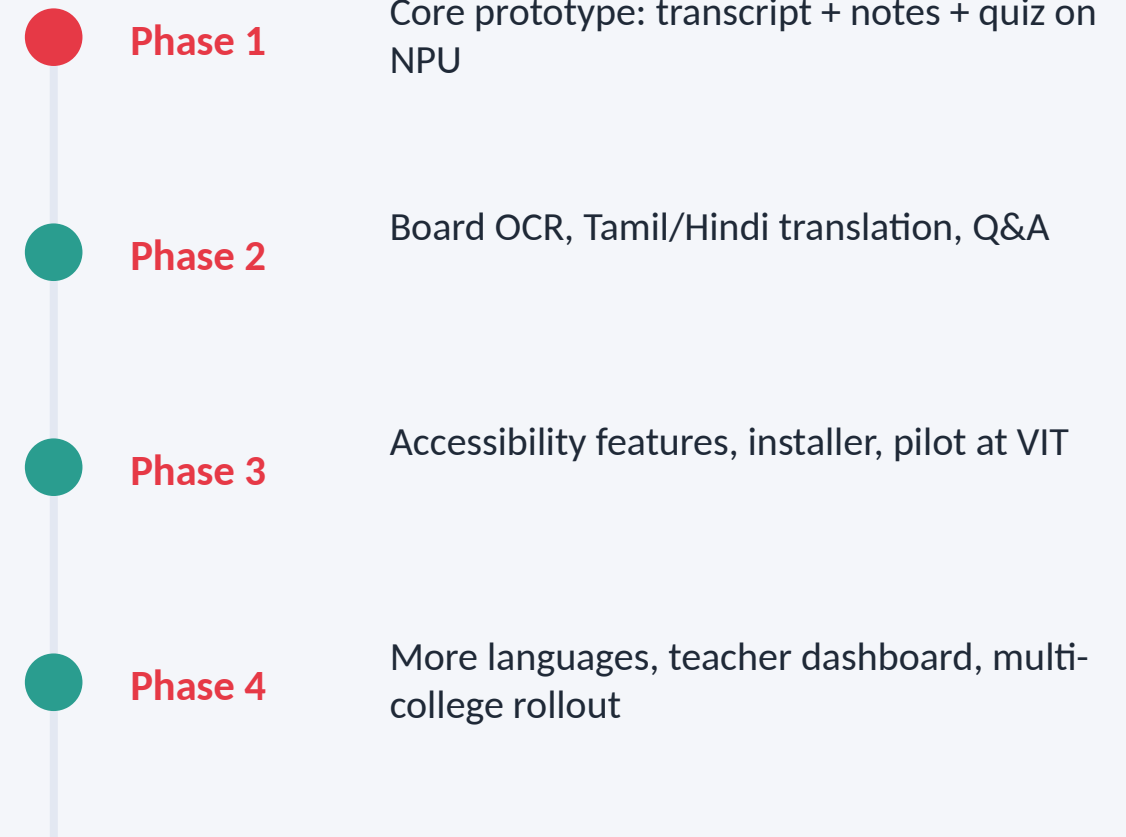
Code on GitHub so colleges and developers can extend it.

Impact & roadmap

From a personal study tool to a campus-wide learning companion

Impact

- Helps regional-medium and first-generation students keep up in English-taught classes
- Frees attention from note-taking to understanding
- Turns every lecture into revision material automatically
- Keeps student data private and costs nothing to run



Thank you

LectureLens — every lecture, understood. Offline, private, on Snapdragon.

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Source code: GitHub repository linked in the submission

